FRANCESCO BANELLI

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EDUCATION

ETH Zurich *Sept 2024 –May 2027 (expected)*
MSc in Robotics, Systems and Control GPA: 5.83/6.0
Master's thesis at Disney Research, starting Nov 2026

Politecnico di Milano *Sept 2021 –Jul 2024*
BSc in Automation and Control Engineering GPA: 29.35/30
Graduation Grade: 110/110 cum Laude

RESEARCH EXPERIENCE

AI Engineering Intern, Flexion Robotics - Perception Team *Mar 2026 –Oct 2026*
Startup building robot-agnostic intelligence for humanoids. Supervisors: Julian Nubert, Dylan Vogel.

Led a project on proprioceptive state estimation for humanoids, where robustness matters most because it is the fallback when cameras and lidar fail. I developed a hybrid estimator in which learning complements, rather than replaces, the robot's kinematics. After the internship, I am continuing the project as first author of a paper for IEEE RA-L.

- Owned the project end to end, from training in simulation to deployment on hardware.
- Showed that it generalizes to unseen motions, where fully learned methods fail.
- Validated it on the Unitree G1 across terrains, improving on classical estimators.

Student Researcher, ETH Zurich - Raffaello D'Andrea's group *Feb 2025 –Feb 2026*
Semester project and continued research. Supervisor: Antonio Terpin.

Worked on fast and accurate fluid-flow estimation: the feedback needed to train reinforcement-learning policies for active flow control directly in the real world.

- Developed ART of PIV, an agent that selects the best flow-estimation pipeline for each image pair, reducing error by 12–22% over the best single estimator at matched speed. Deployed it on the group's water channel, where it learned from only a few minutes of real recordings.
- Co-developed SynthPix, a GPU-parallel generator of synthetic PIV training data (SoftwareX).
- Led the development of Flow Gym, a JAX toolkit for flow estimation (SoftwareX), including GPU ports of DIS and OpenPIV that run 3× and 350× faster than the CPU originals.
- Co-authored a consensus-ADMM method that fuses parallel flow estimators under smoothness and incompressibility priors, reducing error by up to 20% (CDC 2026).
- Co-developed Goggles, an open-source observability framework for JAX robotics experiments.

Student Researcher, Politecnico di Milano - Nicola Gatti's group *Nov 2023 –Jul 2024*
Supervisor: Pierriccardo Olivieri.

- Studied multi-agent coordination and distributed optimization for drone formation control.
- Implemented distributed MPC and potential-field controllers, comparing speed and stability.

PUBLICATIONS

PIVOT: Reliable Proprioceptive State Estimation for Legged Robots
Francesco Banelli, Dylan Vogel, Julian Nubert *In preparation, IEEE RA-L*

ART of PIV: Agentic Real-Time Particle Image Velocimetry for Active Flow Control
Antonio Terpin*, **Francesco Banelli***, Alan Bonomi*, Raffaello D'Andrea
arXiv preprint; to be submitted to Nature Machine Intelligence

PIV Refinement via Consensus ADMM *CDC 2026*
Alan Bonomi*, **Francesco Banelli***, Antonio Terpin*

Flow Gym: A framework for flow-field quantification methods *SoftwareX*
Francesco Banelli*, Antonio Terpin*, Alan Bonomi, Raffaello D'Andrea

SynthPix: A lightspeed PIV image generator *SoftwareX*
Antonio Terpin*, Alan Bonomi*, **Francesco Banelli**, Raffaello D'Andrea

*Equal contribution

HONORS AND AWARDS

Merit Scholarship for Out-of-Town Students, Politecnico di Milano *2021–2024*
Awarded 3 consecutive years in a competitive ranking

2nd Place Nationally, Matematica & Realtà Modeling Competition (2×) *2019, 2021*
Individual mathematical modeling competition; listed in the Italian National Registry of Excellence

TEACHING

Teaching Assistant, ETH Zurich *Sept 2025 –Present*
Game Theory and Control (2 semesters), Linear System Theory (1 semester)

Peer-to-Peer Tutor, Politecnico di Milano *Feb 2023 –Jun 2024*
Linear Algebra, Calculus I, Physics, Control Systems (6 students)

LEADERSHIP AND ENTREPRENEURSHIP

Visual Odometry Team Lead, AEA - Politecnico di Milano *May 2023 –Dec 2023*
Automation Engineering Association, developing a drone built entirely by students.

- Led a three-student team working on visual odometry and sensor fusion.

Co-founder & CEO, CapaCity *Sept 2022 –Nov 2023*
Startup project helping shared-micromobility operators move from free-floating to station-based fleets.

- Selected among the top 45 projects of Switch2Product, Politecnico di Milano's accelerator.
- Won the Verde e Blu Challenge, pitching the project to a public audience.